ECG Heartbeat Classification on Edge-AI

Project instructions — MIT-BIH Arrhythmia Database, MAX78002 deployment workflow, CRISP-DM methodology

Project option for Data Science and Non-AI programs (only if you have **not** worked on this project before). Alternatively, your group can work on its own project.

- At most 5 members can be in a group; one student cannot join more than one group. PhD students must do the project alone.
- Decide on your group in the 1st week of the course. Join any group (some empty groups have already been created) in Canvas as soon as possible, in the People section.
- You can choose a project provided by the teachers, **or** bring your own group project.
- Students in AI programs (TCAIA, TMDIA) at LTU must do Deep Learning in the provided project, or do a project that is Deep Learning based.
- **Everyone:** you are NOT allowed to do the same project you have done before (or are doing currently) in any course you have studied at LTU.

Project Description

Project Overview

In this project, you will develop a compact machine-learning/deep-learning system for **ECG heartbeat classification**.

The project represents the first, ECG-only stage of a larger wearable cardiac-monitoring problem. **No motion sensor or IMU data will be used.**

The intended deployment platform is the **Analog Devices MAX78002 Edge-AI microcontroller**. However:

> ***No physical MAX78002 development kit will be provided or required.***

Your task is therefore to develop a model that is **compatible with the MAX78002 deployment workflow**, demonstrate software-level quantization and synthesis, and critically evaluate its performance.

The project must be carried out according to the **CRISP-DM methodology**:

1. Business/Application Understanding
2. Data Understanding
3. Data Preparation
4. Modeling
5. Evaluation
6. Deployment

CRISP-DM is iterative. You are expected to return to earlier stages when later experiments reveal problems.

Problem

The goal is to classify ECG heartbeats into clinically meaningful heartbeat categories while satisfying the computational constraints of an Edge-AI device (**maximum model size 2 MB**).

Required classification task

Use the MIT-BIH Arrhythmia Database and perform **four-class heartbeat classification**:

Class	Meaning
N	Normal-type beats
S	Supraventricular ectopic beats
V	Ventricular ectopic beats
F	Fusion beats

Use the standard MIT-BIH annotations to construct the classes.

Rare paced/unclassifiable beats that do not belong to the four required classes may be excluded, but all exclusions must be documented.

Important

> You are **not developing a medical device** and you must not claim that your model is clinically validated.
>
> The purpose is to study the complete data-mining and Edge-AI development process.

Main Dataset

MIT-BIH Arrhythmia Database

The primary dataset contains:

- 48 approximately 30-minute ECG recordings
- 47 subjects
- two ECG channels
- sampling frequency of 360 Hz
- expert beat annotations
- approximately 110,000 beat annotations

For the main experiment, use a **patient/record-independent training and testing strategy**. Do **not** randomly distribute beats from the same recording between training and testing.

Recommended test protocol

Use the commonly used inter-patient MIT-BIH partition.

DS1 — Development set

> 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230

Use DS1 for:

- training
- validation
- model selection
- hyperparameter selection

The internal training/validation split must again be performed by **record**, not randomly by heartbeat.

DS2 — Final test set

> 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234

DS2 must remain untouched until final evaluation.

Robustness Dataset

MIT-BIH Noise Stress Test Database

Real ECG recordings contain noise caused by factors such as:

- baseline wander
- muscle/EMG activity
- electrode movement

The Noise Stress Test Database provides representative ECG noise signals.

Your final system must therefore include a **noise-robustness experiment**.

At minimum, evaluate the model under:

- clean ECG
- moderate noise
- strong noise

Recommended SNR conditions include approximately: **12 dB**, **6 dB** and **0 dB**.

You may add noise from the Noise Stress Test Database to held-out ECG signals.

The noise experiment must never be used to leak information from the final test set into model training.

Edge-AI Constraint

The target platform is the **Analog Devices MAX78002**.

The model should therefore be designed as an embedded/Edge-AI model rather than an unrestricted datacenter model.

Prefer operations supported by the MAX78002 CNN accelerator, particularly:

- 1D convolution
- pooling
- ReLU-type activation
- compact fully connected output layers
- fixed-size inputs

Avoid architectures that cannot reasonably be deployed using the MAX78002 CNN accelerator. Examples of architectures that should generally be avoided for this assignment include unrestricted:

- Transformers
- very large ResNets
- LSTMs/GRUs
- models requiring hundreds of MB of parameters
- unsupported custom operations

A **small 1D CNN** is an appropriate starting point. You may develop a better architecture if you can demonstrate that it remains compatible with the MAX78002 toolchain.

Required Experiments

At minimum, your final project must contain:

1. A simple baseline
2. A compact neural-network model
3. FP32 model results
4. INT8/quantized model results
5. Patient-independent evaluation
6. Per-class evaluation
7. Noise robustness evaluation
8. MAX78002 software synthesis/deployment analysis

You are encouraged to investigate issues such as:

- class imbalance
- normalization
- filtering
- ECG window size
- data augmentation
- loss functions
- model size
- quantization-aware training
- robustness versus model complexity

Submission 1 — CRISP-DM: Problem, Data and Preparation

Submission 1 covers the first three CRISP-DM stages. Your report must use the following sections.

1. Business/Application Understanding

1.1 Application Scenario

Explain:

- What problem are you trying to solve?
- Where could automatic ECG classification be useful?
- Why might inference directly on an Edge-AI device be useful?
- Who could potentially use such a system?

1.2 Data-Mining Objective

Clearly state the machine-learning problem. Complete:

Item	To describe
Input	Describe exactly what the model receives.
Output	Describe the classes predicted by the model.
Type of task	Classification, multiclass classification, etc.

1.3 Project Objectives

List your technical objectives. For example:

- distinguish N/S/V/F heartbeats
- generalize to unseen ECG recordings
- remain robust to ECG noise
- develop a compact model
- make the model compatible with MAX78002

1.4 Success Criteria

Define how you will decide whether the project has succeeded. You must define **measurable** criteria rather than simply stating “high accuracy”. Include criteria for:

Predictive performance

Minority-class performance

Noise robustness

Model complexity

MAX78002 compatibility

1.5 Constraints

Discuss:

- limited Edge-AI memory
- fixed-point/INT8 computation
- class imbalance
- limited number of subjects
- noise
- computational resources
- absence of physical MAX78002 hardware

1.6 Risks and Challenges

Identify at least three important risks. For each:

Risk	Why is it important?	Proposed mitigation

2. Data Understanding

2.1 Dataset Description

Report:

- database name
- number of records
- number of subjects
- number of channels
- sampling frequency
- recording duration
- annotation type
- ECG leads available

2.2 Target Classes

Create a table containing:

Final class	Original annotations included	Number of beats
N		
S		
V		
F		

Explain any annotations that you excluded.

2.3 Exploratory Data Analysis

At minimum, show:

- example ECG signal
- example heartbeat from each class
- class distribution
- subject/record distribution
- signal amplitude distribution
- examples of noisy ECG

2.4 Data Quality

Investigate:

- missing data
- extreme amplitudes
- corrupted segments
- class imbalance
- differences between recordings
- differences between ECG channels
- possible noisy regions

2.5 Leakage Analysis

Explain how data leakage could occur in this project. Specifically explain why:

> *Randomly splitting individual heartbeats from the same subject into training and test sets can produce misleading results.*

State exactly how your approach prevents this.

3. Data Preparation

3.1 Record Selection

List exactly which records are used for:

- training
- validation
- testing

3.2 ECG Channel Selection

State:

- which ECG channel/lead you use
- why you selected it
- what you do if the same lead is not available in a recording

3.3 Beat Extraction

Describe:

- how R-peak annotations are obtained
- samples before the annotated beat
- samples after the annotated beat
- final input length

Your network must receive a **fixed-size input**.

3.4 Signal Preprocessing

State exactly which operations are performed and in which order. For example:

```
Raw ECG → filtering → segmentation → normalization → clipping → model input
```

For every preprocessing operation explain **why it is necessary**.

3.5 Label Preparation

Document the complete mapping: original MIT-BIH annotation → final N/S/V/F class.

3.6 Class Imbalance

Report the imbalance. Explain whether you will use:

- class weights
- oversampling
- undersampling
- balanced batches
- focal loss
- another strategy
- no balancing

Justify your decision.

3.7 Data Augmentation

Describe any augmentation used during training. Possible examples include:

- amplitude scaling
- small temporal shifts
- ECG noise
- baseline wander

Augmentation must **only** be applied to the training data.

3.8 Final Dataset

Report:

Split	Subjects/records	N	S	V	F	Total
Train						
Validation						
Test						

3.9 Reproducibility

State:

- software versions
- random seed
- preprocessing parameters
- sampling frequency
- input size

Submission 1 Deliverables

Submit:

1. **CRISP-DM Report — Stages 1–3**
2. **EDA notebook**
3. **Data-preparation code**
4. **Generated train/validation/test statistics**
5. **README** describing how to reproduce the preprocessing

No final neural-network results are required at this stage.

Submission 2 — CRISP-DM: Modeling, Evaluation and Deployment

Submission 2 completes the project. You may revise your conclusions from Submission 1 if your experiments reveal problems. Document significant changes and explain why they were made.

4. Modeling

4.1 Baseline

Implement at least one simple baseline. Examples:

- Logistic Regression
- Random Forest
- SVM
- CNN
- very small neural network

Report its performance.

4.2 Proposed Edge Model

Describe your final neural-network architecture. Provide a table:

Layer	Input	Operation	Output	Parameters

Explain why this architecture is appropriate for Edge-AI.

4.3 Training Configuration

Report:

- optimizer
- learning rate
- batch size
- epochs
- loss function
- early stopping
- class weighting
- augmentation
- random seed

4.4 Model Selection

Explain:

- which models you tried
- which hyperparameters were varied
- which validation metric selected the final model
- why the final architecture was selected

4.5 Quantization

Compare the **FP32 model** versus the **INT8/quantized model**. If Quantization-Aware Training (QAT) is used, explain:

- why QAT is needed
- how it was performed
- whether quantization changed model performance

5. Evaluation

Do not report accuracy alone. At minimum report:

- Accuracy
- Macro F1-score
- Per-class Precision
- Per-class Recall/Sensitivity
- Per-class F1-score
- Confusion matrix

5.1 Clean Test Results

Metric	Baseline	FP32 Edge Model	Quantized Model
Accuracy			
Macro F1			
Model parameters			

5.2 Per-Class Performance

Class	Precision	Sensitivity/Recall	F1
N			
S			
V			
F			

Discuss which classes are difficult and why.

5.3 Quantization Analysis

FP32 Macro F1: _____ INT8 Macro F1: _____ Difference: _____

Discuss the effect of quantization.

5.4 Noise Robustness

Evaluate the final model under progressively increasing ECG noise.

Condition	Accuracy	Macro F1
Clean		
12 dB		
6 dB		
0 dB		

Plot performance versus noise level. Discuss:

- which classes are most affected
- whether degradation is gradual
- whether preprocessing helps
- whether robustness could be improved

5.5 Error Analysis

Select representative errors and investigate them. Include examples such as:

- N classified as S
- S classified as N
- V classified incorrectly
- errors under strong noise

Do not only show numerical results. **Examine the ECG waveforms.**

5.6 Success-Criteria Review

Return to the success criteria defined in Submission 1. For every criterion report **Achieved / Partially achieved / Not achieved** and justify your conclusion.

6. Deployment

For this course, “deployment” means **software-level preparation for deployment to the MAX78002**. A physical development kit is not required.

6.1 MAX78002 Compatibility

Explain:

- why the architecture can run on the MAX78002 CNN accelerator
- which layer types are used
- input dimensions
- output dimensions
- model parameter count

6.2 MAX78002 Toolchain

Use the Analog Devices **ai8x-training** and **ai8x-synthesis** frameworks. Attempt to:

1. prepare the model using the MAX78002-compatible training framework
2. quantize the model
3. synthesize the model for MAX78002
4. generate the device-specific output/C code

6.3 Deployment Evidence

Provide evidence of the synthesis process. Report:

Requirement	Result
Model accepted by MAX78002 toolchain	PASS / FAIL
Quantization completed	PASS / FAIL
Synthesis completed	PASS / FAIL
C/device files generated	PASS / FAIL
Weight-memory requirement	
Data-memory requirement	

If synthesis fails, identify exactly why and modify the architecture if possible.

> A model that obtains excellent accuracy but **cannot be synthesized for the MAX78002 does not fully satisfy the project objective**.

6.4 Hardware Limitations

Because no physical board is supplied, you must **not claim measured hardware values** for:

- inference latency
- power consumption
- energy per inference
- real MAX78002 accuracy
- real-time performance

Clearly distinguish **Software verified** from **Hardware verified**.

6.5 Proposed Hardware Validation

Briefly explain how you would test the model if a MAX78002 kit became available. Include:

- loading the generated model
- supplying identical ECG test inputs
- comparing Python and hardware outputs
- checking intermediate/final output consistency
- measuring inference time
- measuring energy consumption

7. Final Discussion

Discuss:

- What did you learn from the data?
- What was the largest source of error?
- Was the highest-accuracy model also the best Edge-AI model?
- How much performance was lost after quantization?
- How sensitive was the system to ECG noise?
- What compromises were necessary because of hardware constraints?
- What would you change with more time?

8. Limitations

At minimum discuss:

- limited dataset size
- class imbalance
- differences between patients
- use of historical ECG datasets
- absence of prospective clinical testing
- absence of hardware testing
- possible domain shift to wearable ECG devices

9. Final Submission Package

If not using Python, submit an equivalent structure with your files. Submit one compressed project folder containing:

```
GroupXX_ECG_Project/
│
├── report.pdf
├── README.md
├── requirements.txt
│
├── notebooks/
│   └── EDA.ipynb
│
├── src/
│   ├── prepare_data.py
│   ├── dataset.py
│   ├── model.py
│   ├── train.py
│   ├── evaluate.py
│   └── noise_test.py
│
├── configs/
│
├── results/
│   ├── metrics.csv
│   ├── confusion_matrix.*
│   └── noise_results.*
│
└── max78002/
    ├── model_configuration
    ├── quantization_output
    └── synthesis_output
```

Do **not** submit the complete raw PhysioNet datasets. Your README must provide instructions for downloading the required data.

10. Recommended Project Resources

Students should use the following resources supplied on the course page.

Required

- MIT-BIH Arrhythmia Database — PhysioNet
- MIT-BIH Noise Stress Test Database — PhysioNet
- WFDB Python package
- MAX78002 product documentation
- Analog Devices MAX78000/MAX78002 AI documentation
- Analog Devices ai8x-training repository
- Analog Devices ai8x-synthesis repository
- CRISP-DM reference material

Optional

- European ST-T Database
- additional published literature on ECG classification and Edge-AI

Important Rules

1. **Do not randomly mix beats from the same ECG recording across training and testing.**
2. **Do not use the final DS2 test set for model selection.**
3. **Do not optimize hyperparameters using test-set performance.**
4. **Do not report accuracy alone.**
5. **Do not claim clinical validation.**
6. **Do not claim measurements from MAX78002 hardware when no board was used.**
7. **Your submitted model must be designed for MAX78002 compatibility.**
8. **All preprocessing, exclusions and class mappings must be reproducible.**
9. **The project must follow and explicitly document all six CRISP-DM stages.**